NCBI Bookshelf. A service of the National Library of Medicine, National Institutes of Health.

StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-.

# Anesthesia for Awake Craniotomy

**Authors**

Jawdat Alali[1]; Anterpreet Dua[2].

**Affiliations**

[1] Hamad Medical Corporation, Qatar
[2] Medical College of Georgia

Last Update: October 2, 2025.

## Continuing Education Activity

Awake craniotomy requires a carefully tailored anesthetic strategy that balances patient comfort with the need for accurate neurological assessment. Techniques such as monitored anesthesia care and the sleep–awake–sleep approach enable spontaneous ventilation and allow intraoperative cortical mapping while minimizing risks associated with deep sedation or general anesthesia. The anesthetic plan must ensure smooth transitions between sedation and wakefulness, optimize pain control, and avoid interference with electrocorticography. Effective management demands meticulous airway planning, titration of sedatives and analgesics, and close communication among the neurosurgical and anesthesia teams to preserve neurological function while maximizing tumor resection and patient safety.

By participating in this activity, clinicians comprehensively understand anesthetic principles, techniques, and perioperative considerations for awake craniotomy. They enhance their ability to select appropriate candidates, design individualized anesthetic plans, and manage the unique challenges of airway control, sedation, and patient cooperation. The activity strengthens interprofessional communication and team coordination skills, improves patient outcomes, minimizes complications, and ensures the safe and effective integration of awake craniotomy techniques into neurosurgical practice.

**Objectives:**

- Identify early indicators of airway obstruction, hypoventilation, or excessive sedation requiring prompt intervention.
- Assess patient suitability and preoperative factors to optimize safety and cooperation during awake craniotomy.
- Apply appropriate anesthesia and sedation strategies, including scalp blocks and monitored anesthesia care, for indicated awake craniotomy procedures to ensure patient comfort and effective intraoperative mapping.
- Coordinate interprofessional team interventions to minimize perioperative complications and improve patient outcomes.

Access free multiple choice questions on this topic.

## Introduction

Awake craniotomy is a specialized neurosurgical procedure in which the patient is deliberately kept awake during the entire operation or specific portions. The technique does not require the patient to remain fully conscious throughout the surgery; instead, sedation or anesthesia is usually administered during the more painful or surgically stimulating phases, such as application of Mayfield pins, skin incision, craniotomy (bone flap removal), and dural opening. Because brain tissue lacks pain receptors, patients can be fully awake during cortical mapping and lesion resection, crucial stages for preserving neurological function. During these phases, the patient can perform speech or motor tasks in response to intraoperative commands, while the head remains immobilized in a fixed position. Patients are typically maintained under minimal to moderate sedation for the remainder of the surgery.[1]

Awake craniotomy was first performed by Sir Victor Horsley in 1886 to localize the epileptic focus with cortical electrical stimulation.[2] Now, awake craniotomy is most commonly used to map and resect the tumor in vitally important brain areas like the motor and language cortex, where imaging is not sufficiently sensitive.[3] The primary goal is maximal tumor removal while preserving functional brain tissue. During cortical mapping, the awake patient provides real-time feedback when functional areas are stimulated, enabling the surgeon to avoid resecting critical regions. The mapping process involves direct electrical stimulation of the cortex while the patient remains awake and performs relevant tasks, thereby identifying cortical areas where stimulation disrupts the function being tested.[4] With the integration of intraoperative neurophysiologic monitoring and cortical mapping, modern awake craniotomy has become the gold standard for tumors near eloquent areas, offering greater tumor resection, fewer postoperative neurological deficits, and improved survival compared with craniotomy under general anesthesia.[5]

Beyond tumor surgery, awake craniotomy plays a significant role in the management of refractory epilepsy. Intraoperative mapping is essential for identifying seizure foci that often lie adjacent to eloquent areas. Performing the procedure with the patient awake minimizes the suppressive effects of anesthetic agents on cortical recordings.[6] Similarly, language and motor mapping have been successfully employed during resection of vascular lesions, such as arteriovenous malformations, located in functionally critical brain regions. Thus, awake craniotomy has become an invaluable neurosurgical technique, offering both therapeutic efficacy and functional preservation.[7]

## Anatomy and Physiology

Anesthesia for awake craniotomy requires a thorough understanding of cranial anatomy, airway physiology, cerebral physiology, and pain pathways.

### Scalp and Skull

The scalp has 5 layers: skin, connective tissue, galea aponeurotica, loose areolar tissue, and pericranium. The superficial layers are highly vascularized and innervated, accounting for bleeding and nociception during incision. The galea connects the frontalis and occipitalis muscles, while the loose areolar tissue provides a surgical plane for flap elevation. The pericranium, serving as periosteum, nourishes the bone and contributes to periosteal pain.[8]

### Pain Pathways

Nociceptive input originates mainly from the scalp, innervated by branches of the trigeminal (supraorbital, supratrochlear, zygomaticotemporal, auriculotemporal) and cervical nerves (greater, lesser, and least occipital).[9] The periosteum and dura are pain-sensitive, with innervation from meningeal branches of the trigeminal and vagus nerves. In contrast, the calvarial bone, pia mater, and brain parenchyma lack nociceptors, permitting cortical and subcortical stimulation without pain.[10]

### Airway and Respiratory Physiology

Sedation must preserve spontaneous ventilation. Knowledge of airway anatomy is crucial, as airway compromise must be avoided to prevent complications. Sedatives such as propofol and dexmedetomidine reduce respiratory drive; hypercapnia can elevate intracranial pressure (ICP), necessitating close ventilation monitoring.

### Cerebral and Neurophysiology

Awake craniotomy exposes eloquent cortical areas—motor, sensory, and language—that require intraoperative mapping to preserve neurological function. Cerebral blood flow (CBF) is normally autoregulated between mean arterial pressures of 60 to 150 mm Hg, but hypotension and hypertension can compromise perfusion. Sedation, positioning, and PaCO2 directly affect CBF and ICP; hypercapnia induces vasodilation and raises ICP. Direct electrical stimulation (DES) is the cornerstone of awake mapping, enabling real-time functional assessment of language and sensorimotor pathways. This approach allows maximal safe tumor resection while minimizing postoperative neurological morbidity.[11]

## Indications

The following are indications for awake craniotomy:

- **Intrinsic brain tumors**
  - Awake craniotomy is primarily indicated for resection of tumors in eloquent brain regions, allowing real-time functional mapping to maximize tumor removal while minimizing neurological deficits. The cortical localization of temporal lobe language sites in patients with gliomas has established direct cortical stimulation in conscious patients as a reliable method for preserving language function, thereby shaping modern protocols for balancing tumor resection with functional preservation.[12][4] Today, direct electrical stimulation during awake craniotomy is considered the gold standard for safeguarding language and cognitive function during the resection of complex tumors.[13]
- **Epilepsy**
  - Awake craniotomy is indicated in patients with intractable partial epilepsy who are candidates for surgery, as it allows precise localization and resection of the seizure focus using electrocorticography while minimizing the effects of anesthetic agents. This approach is particularly important when the epileptogenic zone is located near or overlaps eloquent cortex, enabling maximal removal of pathological tissue while preserving essential neurological functions.[14][15]
- **Movement disorders**
  - Deep-brain stimulation targeting the subthalamic nucleus or globus pallidus is an established therapy for Parkinson disease, particularly in patients with inadequate symptom control or intolerable motor complications from levodopa. Awake craniotomy is integral to this procedure, as it enables patients to remain responsive during surgery, allowing for real-time mapping and functional testing. This intraoperative feedback ensures accurate electrode placement within critical brain circuits, leading to precise modulation of abnormal neuronal activity. As a result, deep-brain stimulation can significantly improve motor function, reduce disabling symptoms, and enhance quality of life while minimizing the risk of neurological complications.[16][17]
- **Stereotactic procedures**
  - Awake craniotomy can be safely applied for stereotactic brain biopsy, especially in high-risk individuals with multiple comorbidities. By keeping the patient conscious under a scalp block and light sedation, awake craniotomy allows real-time monitoring of neurological function while avoiding the risks associated with general anesthesia.[18]
- **Interventional pain procedures** - Awake craniotomy can be safely applied for functional neurosurgical interventions such as pallidotomy and thalamotomy.[19]

### Benefits of Awake Craniotomy

Awake craniotomy offers multiple advantages over craniotomy performed under general anesthesia. By allowing real-time neurological monitoring during surgery, it maximizes the extent of tumor resection while preserving critical neurological functions such as speech and motor control.[4] Patients typically avoid the risks associated with general anesthesia, including intubation and mechanical ventilation, and experience reduced postoperative pain, nausea, and vomiting.[18][20] The technique also minimizes the need for invasive monitoring—such as arterial lines or catheters—and often eliminates the requirement for postoperative intensive care unit (ICU) monitoring, leading to shorter ICU stays and overall hospitalization.[21][22]

Clinical outcomes are consistently superior, with significantly fewer postoperative neurological deficits (7% vs 23%) and a reduced average hospital stay (1.7 vs 9 days) compared to conventional craniotomy under general anesthesia.[23] These benefits collectively enhance patient recovery, lower long-term healthcare costs, and improve functional outcomes, including preservation of the ability to return to work.[24]

## Contraindications

### Absolute Contraindications

- Patient refusal or inability to cooperate and follow commands, essential for assessing language, memory, and motor functions [18][24]
- Significant confusion, cognitive impairment, or preoperative dysphasia that limits communication [25]
- Inability to remain still during surgery (eg, severe orthopnea)
- Requirement for prone positioning

### Relative Contraindications

- Morbid obesity
- History of obstructive sleep apnea
- Difficult airway or anticipated difficult intubation
- Chronic cough [24]
- Anxiety disorders [18]
- Somnolence [18]
- Marked dysphagia [18]
- High comorbidity burden or poor communication skills
- Poorly controlled seizures
- Needle phobia (trypanophobia) or very low pain threshold
- Surgical cases that are expected to result in major blood loss [18]

## Equipment

Awake craniotomy is a complex procedure that requires specialized equipment to ensure patient safety, comfort, and effective intraoperative mapping. Essential equipment includes:

- Airway management tools
  - Nasal cannula, face mask, nasopharyngeal airways, laryngeal mask airways, and video laryngoscopes should be available for both routine oxygen delivery and emergency airway interventions.
- Monitoring devices
  - Standard American Society of Anesthesiologists (ASA) monitors (electrocardiogram [ECG], pulse oximetry, capnography, temperature, and blood pressure), an invasive arterial line for continuous blood pressure monitoring, and processed electroencephalogram (EEG) devices to titrate sedation [1]
- Infusion and drug delivery systems
  - Programmable infusion pumps for precise titration of sedatives (propofol, dexmedetomidine, remifentanil)
- Regional anesthesia equipment
  - Needles and syringes for scalp block, along with ultrasound guidance if available.
- Temperature management systems
  - Forced-air warming blankets and a temperature-adjustable operating room environment
- Seizure management tools
  - Cold crystalloid irrigation, suction, and rapid access to anticonvulsants and propofol for seizure control
- Positioning and comfort aids
  - Padded headrests, cushions, and restraints to maintain positioning without causing pain or pressure injuries [9]
- Emergency equipment
  - A full range of airway equipment, defibrillators, and resuscitation drugs must be immediately available.

Awake craniotomy is a multidisciplinary process that relies on intraoperative brain mapping, imaging, neuromonitoring, the use of appropriate anesthetic agents, and well-trained teams.[15]

## Personnel

Successful awake craniotomy requires a highly coordinated, multidisciplinary team. Anesthesiologists play a central role in balancing sedation, analgesia, and airway safety while maintaining patient cooperation for functional testing. Neurosurgeons perform tumor resection and cortical mapping, supported by neurophysiologists who conduct intraoperative monitoring, including somatosensory evoked potentials, motor evoked potentials, and electrocorticography.

Neurologists and neuropsychiatrists contribute to preoperative assessment and intraoperative cognitive or behavioral evaluation, while speech therapists are essential for real-time language testing during awake mapping. Specialized nurses are critical throughout all phases of care—they assist with preoperative preparation, intraoperative monitoring, patient positioning, comfort, reassurance, and rapid recognition and management of complications. This team-based approach ensures maximal tumor resection with minimal neurological morbidity.[15]

## Preparation

### Preparation for Anesthesia in Awake Craniotomy

Successful awake craniotomy relies on meticulous preoperative preparation and strong patient rapport. A detailed preoperative interview, conducted with compassionate communication, helps build trust, reduce anxiety, and empower patients by enhancing their sense of control and alleviating distress during the awake phase of their care. The perioperative team—including the surgeon, anesthesiologist, and nurse—should introduce themselves and clearly explain the procedure, positioning, catheter placement, and anticipated intraoperative noises. Patients should be informed about prolonged immobility in a head-fixed position, potential discomfort from urinary catheterization, and noise from the craniectomy, as prolonged immobility is often perceived as more stressful than pain. Patients should also rehearse mapping-related tasks with neurophysiologists before surgery. Gathering personal details for casual conversation can provide additional comfort while providing ongoing reassurance and empathy, and emphasizing the anesthesiologist's presence throughout the procedure further enhances patient confidence, cooperation, and overall experience.[1][24]

### Preoperative Evaluation

Ideal candidates for awake craniotomy are well-motivated, mature patients who can tolerate lying still for several hours and cooperate during intraoperative testing. Thorough preoperative counseling is essential to build trust and reduce anxiety. The anesthesiologist should explain the rationale for the procedure, the expected level of discomfort, the intraoperative tasks, and the potential risks associated with the procedure. Psychological preparation is as important as medical optimization to ensure patient cooperation and comfort.[24][26]

### Premedications

Premedication strategies emphasize seizure management and individualized sedation. Routine anticonvulsant prophylaxis is controversial, as the risk of intraoperative seizures is generally low; however, patients with a history of preoperative seizures remain at higher risk. Intraoperative seizure control may include electrocorticographic monitoring, irrigation of brain tissue with cold crystalloid solution, or small doses of propofol if needed. Premedication should also be tailored to the patient's anxiety, neurological status, comorbidities, and anesthesia plan. Usual medications, such as steroids and antihypertensives, should be continued. Institutional practices differ: some centers routinely administer anticonvulsants and corticosteroids, while others—especially in epilepsy surgery—may withhold anticonvulsants, benzodiazepines, and other agents that suppress epileptiform activity to optimize cortical mapping. Drugs that impair cognition or cause confusion (eg, midazolam, fentanyl, atropine, scopolamine) are generally avoided. However, small doses of midazolam may be used for highly anxious younger patients with preserved neurological function.[24][26]

### Monitoring

Comprehensive intraoperative monitoring is essential for patient safety during awake craniotomy. Standard ASA monitors should be applied, including ECG, pulse oximetry, noninvasive blood pressure, end-tidal CO2 (ETCO2), and temperature. Invasive arterial lines are often preferred over noninvasive blood pressure monitoring to provide continuous measurements and arterial blood gas analysis while avoiding discomfort from repeated cuff inflations. Central venous access may be required for fluid resuscitation or vasoactive drug administration. A precordial Doppler probe should be used to detect venous air embolism for procedures performed in the sitting position.[1][24]

Monitor and catheter placement should be planned carefully to avoid interfering with sensorimotor testing of the limb contralateral to the brain lesion. ETCO2 can be monitored via a modified nasal cannula or the breathing circuit, and respiratory rate may also be estimated from ECG variations. Urinary catheters can be placed under light intravenous sedation or with intraurethral lidocaine to reduce discomfort.[24] Processed EEG monitoring, such as the bispectral index, may be employed to titrate anesthetic depth, particularly during propofol infusion, and to facilitate rapid awakening for intraoperative language and cognitive testing.[24] Temperature control is also crucial: the operating room should be maintained at a comfortable range, and warming blankets can help prevent shivering and anxiety. However, care must be taken to avoid overheating.[1]

## Technique or Treatment

Airway strategies in awake craniotomy vary according to institutional practice and patient factors. The traditional sleep–awake–sleep (SAS) technique with endotracheal intubation ensures secure ventilation but is laborious, complicates reintubation, and may interfere with intraoperative testing. Alternatives include the laryngeal mask airway (LMA), which can provide airway control but may cause hypercapnia from malposition or leaks. Nasopharyngeal airways connected to the breathing circuit are another option for spontaneously breathing patients, though they limit positive-pressure ventilation. An increasingly used approach avoids an endotracheal tube (ETT) and an LMA, relying instead on light to moderate sedation with nasal cannula oxygen and ETCO2 monitoring. While less invasive, this carries risks of obstruction and hypoventilation. No technique has proven superior; the choice depends on patient factors and clinician expertise.[24] Awake craniotomy demands carefully selected anesthetics, sedatives, and analgesics that provide adequate comfort, allow rapid and predictable emergence, minimize respiratory depression, and preserve effective communication between the patient and the surgical team.[27]

### Anesthesia Technique

The choice of anesthetic technique for awake craniotomy depends on patient factors, surgical complexity, and institutional preference. Two main approaches are commonly used: monitored anesthesia care (MAC) and the SAS technique, often supplemented with regional anesthesia, such as a scalp block, primarily for analgesia.[24]

### MAC

In MAC, patients breathe spontaneously under moderate sedation, with sedative depth titrated to the surgical phase. Sedation is increased during painful steps—such as pinning, skin incision, and bone flap removal—reduced or paused during cortical mapping, and resumed for closure. This approach emphasizes using low-dose sedatives to preserve spontaneous ventilation and ensure a smooth asleep–awake transition. Common agents include low-dose propofol (20–150 µg/kg/min), remifentanil (0.01–0.06 µg/kg/min, often via target-controlled infusion), and dexmedetomidine (0.5–1 µg/kg bolus followed by 0.3–1 µg/kg/h), which provides sedation without respiratory depression and minimally affects electrocorticography. However, it can cause hypotension and bradycardia.[9] Remimazolam, an ultrashort-acting benzodiazepine reversible with flumazenil, permits rapid awakening, whereas other benzodiazepines are generally avoided for their interference with electrocorticography.

The primary goals are to minimize sedative exposure to reduce delirium risk and to maintain optimal conditions for reliable neuropsychological testing. Advantages include a 20- to 30-minute reduction in surgical time, avoidance of risks associated with deep sedation, and excellent efficacy when performed by experienced teams. Disadvantages include a higher incidence of intraoperative agitation, seizures, desaturation episodes, pain complaints, and a slight increase in brain edema. MAC is typically reserved for high-volume centers with well-coordinated teams; its role in high-risk patients remains uncertain.[28]

### SAS

SAS begins with general anesthesia and airway management induction using either endotracheal intubation or an LMA. The patient remains anesthetized during pinning, craniotomy, and dural opening, is awakened for cortical mapping, and then reanesthetized for surgical closure. Endotracheal intubation provides secure ventilation but can complicate awakening through coughing or agitation, potentially leading to increased brain swelling. In contrast, an LMA is often preferred for smoother transitions, despite a risk of displacement. Total intravenous anesthesia with propofol and remifentanil is standard, with dexmedetomidine sometimes added; in some cases, sevoflurane is used during sleep phases to facilitate rapid transitions. SAS generally involves deeper sedation with higher anesthetic doses and mechanical ventilation—often via LMA—before the awake phase.[28]

The goals are to improve patient and surgical team comfort, prevent pain, hypoventilation, or intraoperative movement, control brain swelling through hyperventilation, and minimize patient awareness during the awake stage. Advantages include a secure airway, reliable sedation, and improved control of physiologic variables. Disadvantages include longer preparation times and an increased incidence of hypertensive episodes. Despite these limitations, surgical outcomes are comparable to MAC.[9][28] SAS is often selected for patients without significant comorbidities or aphasia and in centers early in their awake craniotomy experience.

### Regional Scalp Block

Intraoperative pain perception during awake craniotomy procedures primarily originates from soft tissues and pericranial muscles. If uncontrolled, it can trigger increased catecholamine release and oxygen consumption, leading to brain hyperemia, elevated intracranial pressure, and potential complications such as intracranial hematoma. Therefore, effective analgesia is essential for patient comfort, reducing systemic stress responses, and facilitating early rehabilitation.[29] Scalp blocks have been shown to lower postoperative pain scores, reduce opioid requirements, and minimize opioid-related adverse effects such as sedation, which could mask neurological assessment. Early analgesia may also prevent central sensitization and reduce the risk of chronic pain.[29]

Scalp blocks are widely used to mitigate hemodynamic responses and incisional pain; however, the optimal timing and standardized protocols for their use remain under investigation.[29] Bilateral scalp blocks are commonly performed before pinning the head in Mayfield pins to provide effective analgesia during awake craniotomy. This technique targets the sensory nerves of the scalp—including the supraorbital and supratrochlear nerves (V1), zygomaticotemporal and auriculotemporal nerves (V2), and the greater and lesser occipital nerves (C2–C3)—helping maintain hemodynamic stability, attenuate the sympathetic response to painful stimuli, and improve patient tolerance, particularly during the awake phase for cortical mapping.[9][25]

At many institutions, anesthesiologists perform blocks at pinning sites, while surgeons administer blocks along the incision line, which is marked in advance to guide nerve selection. A bolus of fentanyl (50–100 µg) before needle insertion can improve patient comfort.[9] Local anesthetic volumes of 2 to 5 mL per injection site typically produce a 1 cm subcutaneous wheal sufficient for analgesia. Ultrasound guidance is increasingly employed to avoid intracerebral injection near bony defects, particularly in redo craniotomies.

### Block Technique

- **Supraorbital nerve**
  - This nerve is blocked at the supraorbital notch above the eyebrow with the needle inserted perpendicularly. Care is taken to palpate the orbital rim to avoid injury to the eyeball.
- **Supratrochlear nerve**
  - This nerve is blocked approximately 2 cm from the supraorbital nerve at the orbital rim.
- **Auriculotemporal nerve**
  - This nerve is blocked about 1 cm anterior to the tragus at a depth of 0.5 to 1 cm to minimize the risk of facial nerve involvement.
- **Zygomaticotemporal nerve**
  - This nerve pierces the deep temporalis fascia near the frontozygomatic suture and has variable branching. Local anesthetics can be injected at multiple levels—deep near the periosteum and superficial in the fascia and subcutaneous tissue. Alternatively, a subcutaneous ring block extending from the lateral eyebrow to the anterior tragus may be used.
- **Greater occipital nerve**
  - This nerve is blocked medial to the occipital artery along a line connecting the occipital protuberance and mastoid process, with caution to avoid intraarterial injection.
- **Lesser occipital nerve**
  - This nerve is blocked at the midpoint between the greater occipital nerve and the mastoid along the superior nuchal line.
- **Greater auricular nerve**
  - This nerve is blocked posterior to the ear.

A mixture of long-acting (ropivacaine or levobupivacaine) and short-acting (lidocaine) local anesthetics with epinephrine is typically used, with 3 to 4 mL per nerve (except the zygomaticotemporal nerve), prepared as a 60 mL mixture at our institution. Additional infiltration at planned incision sites is often performed to ensure adequate analgesia.[9][25]

## Complications

Challenges and complications during the awake phase include:

- **Seizures**
  - The incidence of seizures during awake craniotomy ranges from 2% to 22%.[24][25][30][31] Seizures most commonly occur during cortical or subcortical stimulation for brain mapping. They are often focal, brief, and self-limiting, though generalized seizures can also occur.
  - Risk factors include a history of preoperative seizures, younger age, and tumor location, particularly in the frontal lobe or near the motor cortex. Higher stimulation intensity and frequency may further increase the risk.[25][30] Intraoperative seizures are associated with transient motor deterioration and longer hospital stays.[31]
  - First-line management involves irrigation of the cortex with ice-cold sterile saline, repeated as needed. If seizures persist, intravenous propofol (10–50 mg) or midazolam (1–2 mg) can be administered.
  - Most seizures resolve without lasting consequences, though rare cases may result in apnea or cardiac arrest.[24][30]
- **Hypertension**
  - Hypertension is a commonly encountered intraoperative complication, reported in 27% of patients, while hypotension occurs in about 10%.[32][33] This issue is often secondary to pain, agitation, or anxiety, with higher intraoperative anxiety observed in female patients and in those younger than 60.[34] Other causes, such as hypoxia or hypercapnia, should be evaluated and addressed. Temporizing management may include administration of intravenous labetalol or esmolol.
- **Nausea and vomiting**
  - Nausea occurs in approximately 4% of patients undergoing awake craniotomy and is usually related to opioids, anxiety, or surgical stimulation. Management options include ondansetron, dexamethasone, or propofol.[33]
- **Respiratory complications**
  - Airway obstruction is a major concern during awake craniotomy, often resulting from excessive sedation and leading to hypoxia or hypercarbia.[24][32] Management begins with rapid assessment and immediate intervention, including alerting the surgical team, stopping sedative infusions, and providing mask ventilation with 100% oxygen using a jaw thrust and, if needed, oral or nasopharyngeal airways. Assisted ventilation or airway devices such as an LMA, video laryngoscope, or ETT should be readily available. LMA placement can be challenging due to patient positioning, and intubation may require assistance.
  - If apnea or chest rigidity occurs due to remifentanil, stopping the infusion and attempting mask ventilation is recommended; low-dose succinylcholine (0.5 mg/kg) may be used if necessary.[24]
- **Air embolism**
  - Venous air embolism (VAE) can occur in 20% to 40% of craniotomies performed in the sitting position and may lead to intraoperative respiratory distress. Transesophageal echocardiography (TEE) is the most sensitive method for detecting air embolism, capable of identifying as little as 0.02 mL/kg of air; however, it is an invasive procedure. Precordial Doppler is the most commonly used noninvasive monitoring tool for VAE detection.[32]
- **Neurological deficits**
  - Neurological deficits are among the most commonly reported complications of awake craniotomy, including hemiplegia, paresis, dysphasia or aphasia, cranial nerve dysfunction, hemianopia, and memory impairment. Most deficits are transient and improve with follow-up or supportive management, although a minority may persist as long-term impairments.[35]
- **Hyponatremia**
  - Hyponatremia is the most frequent electrolyte imbalance in neurosurgical individuals, most commonly caused by the syndrome of inappropriate antidiuretic hormone secretion (SIADH).[36] Other causes, such as acute adrenocorticotropic hormone deficiency or cerebral salt wasting syndrome (CSWS), should also be considered. SIADH leads to water retention and dilutional hyponatremia in euvolemic individuals, whereas CSWS causes excessive natriuresis and relative hypovolemia.
  - Perioperative factors—including rapid mannitol infusion, medications such as carbamazepine, thiazides, or desmopressin, and preexisting electrolyte disturbances—can contribute to hyponatremia.[36][37]
  - Hyponatremia may increase intracranial pressure, delay awakening, and cause neurological deterioration, ranging from mild attentional or gait disturbances to severe disorientation, seizures, or coma. Careful monitoring of plasma sodium and timely correction, often with intravenous hypertonic saline instead of mannitol, is essential to minimize neurological complications and maintain patient safety during awake craniotomy.[36][37]
- **Bradycardia**
  - Although scalp nerve blocks are generally safe, they may occasionally cause significant hemodynamic disturbances. Sudden bradycardia and hypotension can occur, most often due to the trigeminal cardiac reflex triggered by stimulation of trigeminal nerve branches. Opioid use may increase susceptibility, emphasizing the importance of slow incremental infiltration and vigilant monitoring during awake craniotomy.[38]
- **Failed awake craniotomy**
  - Awake craniotomy is considered a failure if conversion to general anesthesia is required or if adequate mapping or monitoring cannot be achieved. The failure rate is approximately 2% (range, 0%–6%) and can be minimized by selecting patients appropriately.[39] Causes of failure include intraoperative complications such as generalized seizures necessitating conversion to general anesthesia.[32]

### Postoperative Care

After awake craniotomy, patients are typically transferred to a high-dependency or neurointensive care unit for overnight observation. Pain is managed using multimodal analgesia, including regional techniques (scalp block or local infiltration), acetaminophen, and small doses of intravenous opioids, sometimes administered via patient-controlled analgesia. Compared to craniotomy under general anesthesia, awake craniotomy is associated with lower postoperative analgesic requirements and a reduced incidence of nausea and vomiting. Hospital stays are generally shorter, averaging 3 to 4 days, compared to 9 days for craniotomy under general anesthesia, with some patients eligible for same-day discharge. Early postoperative visits allow anesthesia providers to assess patient experiences and short-term outcomes more effectively.[24]

## Clinical Significance

Anesthesia plays a pivotal role in awake craniotomy by balancing patient comfort, safety, and effective intraoperative functional mapping. Tailored anesthesia strategies—including MAC, SAS protocols, and regional scalp blocks—allow patients to remain cooperative during cortical and subcortical mapping while minimizing pain, anxiety, and respiratory compromise. Optimized anesthetic management enhances neurological preservation, reduces postoperative complications, and facilitates rapid recovery. Conversely, inadequate anesthesia may lead to patient agitation, failed mapping, intraoperative seizures, or respiratory compromise, emphasizing the need for skilled planning and interprofessional coordination.

Awake craniotomy with intraoperative brain mapping remains the gold standard for resecting tumors or lesions in or near eloquent brain regions. Direct electrical stimulation of the cortex enables patients to perform relevant tasks, allowing the surgical team to identify areas critical for speech and motor functions and avoid permanent deficits. Careful patient selection and thorough preoperative counseling are essential. Only cooperative individuals who follow commands during surgery are suitable candidates for this procedure. Clear communication about the procedure, expected patient participation, and the use of sedation and nerve blocks ensures patients remain calm, comfortable, and responsive throughout the operation, maximizing surgical success and safety.

## Enhancing Healthcare Team Outcomes

Effective anesthesia management for awake craniotomy requires coordinated interprofessional collaboration to optimize patient-centered care, safety, and surgical outcomes. The anesthesiologist leads sedation, analgesia, and airway management, assisted by nurses and anesthesia technologists who ensure patient comfort, safe positioning, and continuous monitoring of vital signs throughout transitions in sedation. Neurosurgeons focus on the safe resection of tumors or lesions and functional mapping, while neurophysiologists provide intraoperative monitoring to guide the preservation of cortical and subcortical function. Preoperative planning, including patient counseling and selection, aligns team strategies, reduces patient anxiety, and prepares the team for anticipated challenges.

Clear, continuous communication during the procedure ensures rapid response to seizures, hemodynamic changes, or airway compromise. By integrating specialized skills—such as airway management, sedation titration, neurocognitive assessment, and hemodynamic monitoring—while respecting patient autonomy, the interprofessional team minimizes complications, preserves neurological function, and improves procedural outcomes. Flattening traditional hierarchies and fostering shared responsibility empowers all team members, enhances patient cooperation, and strengthens collaborative decision-making across preoperative, intraoperative, and postoperative phases.

## Nursing, Allied Health, and Interprofessional Team Interventions

Successful awake craniotomy relies on coordinated interprofessional teamwork. Preoperative planning between the anesthesiologist and surgical team ensures that patient-specific considerations, required equipment, and optimal positioning are addressed before entering the operating room. Collaboration with operating room nurses, technicians, and other allied health professionals is essential to anticipate procedural needs and ensure patient safety. Clear, continuous communication among anesthesiologists, nurse anesthetists, surgeons, nurses, and technicians minimizes intraoperative complications, supports a smooth workflow, and contributes to optimal surgical outcomes.

## Review Questions

- Access free multiple choice questions on this topic.
- Click here for a simplified version.
- Comment on this article.

## References

1. Min KT. Practical guidance for monitored anesthesia care during awake craniotomy. Anesth Pain Med (Seoul). 2025 Jan;20(1):23-33. [PMC free article: PMC11834892] [PubMed: 39923769]
2. Piccioni F, Fanzio M. Management of anesthesia in awake craniotomy. Minerva Anestesiol. 2008 Jul-Aug;74(7-8):393-408. [PubMed: 18612268]
3. Archer DP, McKenna JM, Morin L, Ravussin P. Conscious-sedation analgesia during craniotomy for intractable epilepsy: a review of 354 consecutive cases. Can J Anaesth. 1988 Jul;35(4):338-44. [PubMed: 3402010]
4. McBenedict B, Hauwanga WN, Fong YB, Pogodina A, Obinna EE, Pradhan S, Kazmi SS, Netto JGM, Lima Pessôa B. Awake Craniotomy in Neurosurgery: A Bibliometric Analysis of the Top 100 Most-Cited Articles and Review of Technological Advancements. Cureus. 2024 Dec;16(12):e76290. [PMC free article: PMC11754922] [PubMed: 39850176]
5. Kim SH, Choi SH. Anesthetic considerations for awake craniotomy. Anesth Pain Med (Seoul). 2020 Jul 31;15(3):269-274. [PMC free article: PMC7713838] [PubMed: 33329824]
6. Sommer B, Grummich P, Coras R, Kasper BS, Blumcke I, Hamer HM, Stefan H, Buchfelder M, Roessler K. Integration of functional neuronavigation and intraoperative MRI in surgery for drug-resistant extratemporal epilepsy close to eloquent brain areas. Neurosurg Focus. 2013 Apr;34(4):E4. [PubMed: 23544410]
7. Gabarrós A, Young WL, McDermott MW, Lawton MT. Language and motor mapping during resection of brain arteriovenous malformations: indications, feasibility, and utility. Neurosurgery. 2011 Mar;68(3):744-52. [PubMed: 21311300]
8. Kemp WJ, Tubbs RS, Cohen-Gadol AA. The innervation of the scalp: A comprehensive review including anatomy, pathology, and neurosurgical correlates. Surg Neurol Int. 2011;2:178. [PMC free article: PMC3262995] [PubMed: 22276233]
9. Min KT. Erratum: Practical guidance for monitored anesthesia care during awake craniotomy. Anesth Pain Med (Seoul). 2025 Apr;20(2):185-186. [PMC free article: PMC12080624] [PubMed: 40350159]
10. Sung CH, Tsuang FY, Lin CP, Chan KC, Chou WH, Wu CY. Influence of scalp block on oncological outcomes of high-grade glioma in adult patients with and without isocitrate dehydrogenase-1 mutation. Sci Rep. 2021 Aug 13;11(1):16489. [PMC free article: PMC8363618] [PubMed: 34389754]
11. Saito T, Tamura M, Chernov MF, Ikuta S, Muragaki Y, Maruyama T. Neurophysiological Monitoring and Awake Craniotomy for Resection of Intracranial Gliomas. Prog Neurol Surg. 2018;30:117-158. [PubMed: 29241172]
12. Haglund MM, Berger MS, Shamseldin M, Lettich E, Ojemann GA. Cortical localization of temporal lobe language sites in patients with gliomas. Neurosurgery. 1994 Apr;34(4):567-76; discussion 576. [PubMed: 7516498]
13. Sherry N, Gecici NN, Stepniak A, Miller D, Habib A, Niranjan A, Barrios-Martinez J, Yeh FC, Balzer J, Zinn PO. Case Report: A personalized model of care for surgical resection of complex brain tumor with atypical language dominance. Front Neurosci. 2025;19:1587594. [PMC free article: PMC12328450] [PubMed: 40778355]
14. Korkar GH, Isnard J, Montavont A, Catenoix H, Rheims S, Guénot M. Awake craniotomy for epilepsy surgery on eloquent speech areas: a single-centre experience. Epileptic Disord. 2021 Apr 01;23(2):347-356. [PubMed: 33926856]
15. Mofatteh M, Mashayekhi MS, Arfaie S, Adeleye AO, Jolayemi EO, Ghomsi NC, Shlobin NA, Morsy AA, Esene IN, Laeke T, Awad AK, Labuschagne JJ, Ruan R, Abebe YN, Jabang JN, Okunlola AI, Barrie U, Lekuya HM, Idi Marcel E, Kabulo KDM, Bankole NDA, Edem IJ, Ikwuegbuenyi CA, Nguembu S, Zolo Y, Bernstein M. Awake Craniotomy in Africa: A Scoping Review of Literature and Proposed Solutions to Tackle Challenges. Neurosurgery. 2023 Aug 01;93(2):274-291. [PMC free article: PMC10319364] [PubMed: 36961213]
16. Deogaonkar A, Avitsian R, Henderson JM, Schubert A. Venous air embolism during deep brain stimulation surgery in an awake supine patient. Stereotact Funct Neurosurg. 2005;83(1):32-5. [PubMed: 15821367]
17. Eiamcharoenwit J, Akavipat P. Incidence of complications associated with deep brain stimulation surgery in patients with Parkinson's disease: An 8-year retrospective study. Saudi J Anaesth. 2024 Jan-Mar;18(1):62-69. [PMC free article: PMC10833010] [PubMed: 38313714]
18. Yazar CO, Sezgi A, Zengin M, Ünlüer C, Kına SF, Arık E, Özdemir Sezgi EŞ, Ergil J. Stereotactic Brain Biopsy with Awake Craniotomy: Our Awake Craniotomy Experience on a Complicated Case and Mini Review. Turk J Anaesthesiol Reanim. 2025 Jul 24;53(4):177-182. [PMC free article: PMC12288426] [PubMed: 40619766]
19. Al Fudhaili AN, Al-Busaidi F, Madan ZM, Al Issa MS, Al Mamria MH, Al-Saadi T. Awake Craniotomy Surgery in Pediatrics: A Systematic Review. World Neurosurg. 2023 Nov;179:82-87. [PubMed: 37595837]
20. Manninen PH, Tan TK. Postoperative nausea and vomiting after craniotomy for tumor surgery: a comparison between awake craniotomy and general anesthesia. J Clin Anesth. 2002 Jun;14(4):279-83. [PubMed: 12088812]
21. Brown T, Shah AH, Bregy A, Shah NH, Thambuswamy M, Barbarite E, Fuhrman T, Komotar RJ. Awake craniotomy for brain tumor resection: the rule rather than the exception? J Neurosurg Anesthesiol. 2013 Jul;25(3):240-7. [PubMed: 23603885]
22. Hervey-Jumper SL, Berger MS. Technical nuances of awake brain tumor surgery and the role of maximum safe resection. J Neurosurg Sci. 2015 Dec;59(4):351-60. [PubMed: 26394241]
23. Groshev A, Padalia D, Patel S, Garcia-Getting R, Sahebjam S, Forsyth PA, Vrionis FD, Etame AB. Clinical outcomes from maximum-safe resection of primary and metastatic brain tumors using awake craniotomy. Clin Neurol Neurosurg. 2017 Jun;157:25-30. [PubMed: 28384595]
24. Meng L, McDonagh DL, Berger MS, Gelb AW. Anesthesia for awake craniotomy: a how-to guide for the occasional practitioner. Can J Anaesth. 2017 May;64(5):517-529. [PubMed: 28181184]
25. Abdelhameed E, Abdelghany MS, Abdelkhalek H, Elatrozy HIS. Awake surgery for lesions near eloquent brain under scalp block and clinical monitoring: experience of single center with limited resources. Egypt J Neurol Psychiatr Neurosurg. 2021;57(1):78. [PMC free article: PMC8205193] [PubMed: 34149281]
26. Izzi A, Mincolelli G, D'Onofrio G, Marchello V, Manuali A, Icolaro N, Mirabella L, Riefolo A, Mazzotta B, Barile A, Gorgoglione LP, Del Gaudio A. Awake Craniotomy in Conscious Sedation: The Role of A2 Agonists. Brain Sci. 2024 Jan 30;14(2) [PMC free article: PMC10886693] [PubMed: 38391722]
27. Viderman D, Nabidollayeva F, Bilotta F, Abdildin YG. Comparison of dexmedetomidine and propofol for sedation in awake craniotomy: A meta-analysis. Clin Neurol Neurosurg. 2023 Mar;226:107623. [PubMed: 36791589]
28. Natalini D, Ganau M, Rosenkranz R, Petrinic T, Fitzgibbon K, Antonelli M, Prisco L. Comparison of the Asleep-Awake-Asleep Technique and Monitored Anesthesia Care During Awake Craniotomy: A Systematic Review and Meta-analysis. J Neurosurg Anesthesiol. 2022 Jan 01;34(1):e1-e13. [PubMed: 31972627]
29. Chen Y, Ni J, Li X, Zhou J, Chen G. Scalp block for postoperative pain after craniotomy: A meta-analysis of randomized control trials. Front Surg. 2022;9:1018511. [PMC free article: PMC9550001] [PubMed: 36225222]
30. Rajan S, Cata JP, Nada E, Weil R, Pal R, Avitsian R. Asleep-awake-asleep craniotomy: a comparison with general anesthesia for resection of supratentorial tumors. J Clin Neurosci. 2013 Aug;20(8):1068-73. [PubMed: 23453156]
31. Nossek E, Matot I, Shahar T, Barzilai O, Rapoport Y, Gonen T, Sela G, Grossman R, Korn A, Hayat D, Ram Z. Intraoperative seizures during awake craniotomy: incidence and consequences: analysis of 477 patients. Neurosurgery. 2013 Jul;73(1):135-40; discussion 140. [PubMed: 23615101]
32. Sokhal N, Rath GP, Chaturvedi A, Dash HH, Bithal PK, Chandra PS. Anaesthesia for awake craniotomy: A retrospective study of 54 cases. Indian J Anaesth. 2015 May;59(5):300-5. [PMC free article: PMC4445152] [PubMed: 26019355]
33. Abaziou T, Tincres F, Mrozek S, Brauge D, Marhar F, Delamarre L, Menut R, Larcher C, Osinski D, Cinotti R, Sol JC, Fourcade O, Roux FE, Geeraerts T. Incidence and predicting factors of perioperative complications during monitored anesthesia care for awake craniotomy. J Clin Anesth. 2020 Sep;64:109811. [PubMed: 32320919]
34. Mofatteh M, Mashayekhi MS, Arfaie S, Chen Y, Hendi K, Kwan ATH, Honarvar F, Solgi A, Liao X, Ashkan K. Stress, Anxiety, and Depression Associated With Awake Craniotomy: A Systematic Review. Neurosurgery. 2023 Feb 01;92(2):225-240. [PMC free article: PMC9815094] [PubMed: 36580643]
35. Tariq R, Siddiqui UA, Bajwa MH, Baig AN, Khan SA, Tariq A, Bakhshi SK. Feasibility of awake craniotomy for brain arteriovenous malformations: A scoping review. World Neurosurg X. 2024 Apr;22:100321. [PMC free article: PMC10911851] [PubMed: 38440377]
36. Yamamoto S, Masaki H, Kamata K, Nomura M, Ozaki M. A case of failed awake craniotomy due to progressive intraoperative hyponatremia. JA Clin Rep. 2018 May 15;4(1):40. [PMC free article: PMC6967330] [PubMed: 32026054]
37. Medeiros H, Nascimento MS, Da Silva Ferreira LP, Rocha T, Da Silva WA. Opioid-Free Anesthesia for Awake Neurosurgery in a Patient With Asthma and Von Willebrand Disease: A Case Report. Cureus. 2023 Oct;15(10):e47103. [PMC free article: PMC10646757] [PubMed: 38022194]
38. Chowdhury T, Baron K, Cappellani RB. Severe bradycardia during scalp nerve block in patient undergoing awake craniotomy. Saudi J Anaesth. 2013 Jul;7(3):356-7. [PMC free article: PMC3757818] [PubMed: 24015148]
39. Nossek E, Matot I, Shahar T, Barzilai O, Rapoport Y, Gonen T, Sela G, Korn A, Hayat D, Ram Z. Failed awake craniotomy: a retrospective analysis in 424 patients undergoing craniotomy for brain tumor. J Neurosurg. 2013 Feb;118(2):243-9. [PubMed: 23121432]

**Disclosure:** Jawdat Alali declares no relevant financial relationships with ineligible companies.

**Disclosure:** Anterpreet Dua declares no relevant financial relationships with ineligible companies.

Copyright © 2025, StatPearls Publishing LLC.

This book is distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits others to distribute the work, provided that the article is not altered or used commercially. You are not required to obtain permission to distribute this article, provided that you credit the author and journal.

Bookshelf ID: NBK572053 PMID: 34283419